

AI adoption by European SMEs

Week 1: welcome, discussion, and the assignments.

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Session of Monday 7 September, about four hours with one break, both lecturers in the room.

Goal: everyone leaves in a team of four, with both first drafts started, and we leave with a read on where the group starts.

Pre-reading was Draghi's address to the European Parliament, eleven pages. Ask who read it; do not make a point of it yet.

Two modules, one question

- **AEL — tech.** Led by Witek ten Hove. Builds the tool.
- **AIBS — research.** Led by Meike Botterweg. Decides what the tool's output must achieve, and judges it.
- **One research problem.** What do European SMEs need to implement AI in their business?

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Two modules, taught as one pipeline to the same teams.

- AEL, the technical track: builds the tool. Witek.
- AIBS, the research track: decides what the tool's output must achieve, and judges whether it does. Meike.

- One research problem holds both: what do European SMEs need in order to implement AI in their business?

Say once: you never work on one module at a time. Every week has both hats.

How this module runs

- **One pipeline, two modules.** AEL builds the tool. AIBS decides what its output must achieve, and judges it.
- **Teams of four, both modules.** Every team works together on the AEL and AIBS assignments; each student also holds one AEL build role.
- **The goal:** a Handbook about the state of AI, aimed at SMEs — what a regional owner needs to know and decide.
- **The rhythm:** one new published page per team, per week.
- **The scope:** 6 weeks of classes, then one combined assessment for both modules.

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Keep this short: all of it is on the course page in writing, and they will read it again.

- The handbook is the end goal — what a regional SME owner needs to know and decide about AI.

- One new published page per team per week. New page, never an edit to an earlier one, so the trail of thinking stays visible.
- Teams of four; everyone researches; each holds one AEL build role.
- Six Mondays, then an assessment period: consolidated handbook and one joint interview, two marks.

We're running this week by week

FIXED

- The end goal — the Handbook
- 6 weeks + assessment
- Teams of four, one build role each
- One page per team per week
- Two informational gates: Socratic + peer
- Final: consolidated handbook + joint interview

DECIDED AS WE GO

- How much scaffolding each assignment needs
- The exact sub-theme and depth per week
- Whether the field visit runs weeks 2–4 as planned
- How far we get through the provisional week map

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We run this adaptively. We start with the first assignments, watch what the group can do, and decide week by week what comes next.

Be explicit about the consequence: some things you want to know in week 1 do not exist yet in week 1. Each week's page appears about a week before its

session.

Left column is a promise — it will not move. Right column is decided as we go, and that is deliberate, so we pitch the work at the level in the room.

Today's session (about 4 hours)

- **Welcome** — how the module runs
- **Discussion 1** — your experience with AI
- **Discussion 2** — the Draghi report
- Break
- **Team formation** — teams of four, one build role each
- **Assignments** — kick off the PRD and the research proposal
- **Share-out** — one sentence per team: who your handbook is for

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Timings, 240 minutes:

- 0:00–0:20 welcome and how the module runs
- 0:20–0:55 discussion 1, your experience with AI
- 0:55–1:35 discussion 2, the Draghi report
- 1:35–1:45 break
- 1:45–2:10 team formation
- 2:10–2:25 the two assignments
- 2:25–3:40 team work time, both lecturers circulating
- 3:40–4:00 share-out and close

Your experience with AI

This is also our diagnostic — it shapes what we do in the coming weeks.

- Which AI tools have you actually used, and for what?
- Where has it clearly helped you? Where has it failed or misled you?
- Confidence check: using AI · building with AI · judging AI output
- How will AI change the job market you're about to enter?
- What skills will you need — which do you have, which are gaps?
- What do you want to be able to do by the end of this minor?

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Structured round; capture the answers on the board. Say that this is our diagnostic: it shapes the coming weeks, so understating what you can do does not help.

Confidence check by hands, 1 to 5, three times: using AI tools; building something with AI; judging whether an AI output is any good.

Keep a written record of three things: the confidence spread, the job-market and skills answers, and the "want to be able to do" answers. They go into the first decision-log entry after class.

The Draghi report

- **What it is.** Mario Draghi's report for the European Commission (September 2024) on why the EU is falling behind the US and China.
- **Core diagnosis.** The gap is largely a technology gap — and Europe is slow to diffuse digital tech into existing businesses, especially SMEs.
- **On AI.** Europe won't win only by building frontier models; it has to get AI adopted across its industrial base.
- **SME barriers.** Fragmented single market, regulatory complexity, limited capital access, skills shortages, limited scale.
- **Why it's our starting point.** The Handbook is a small, concrete contribution to exactly that diffusion problem.

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Framing, about ten minutes, lecturer only.

- What it is: Draghi's report for the European Commission, September 2024, on why the EU falls behind the US and China on growth and productivity.
- Core diagnosis: largely a technology gap. Few large tech firms, and, the point that matters for us, slow to diffuse digital technology into the businesses Europe already has, especially SMEs.
- On AI: Europe will not win by building frontier models alone; it has to get AI adopted across its industrial base.
- Barriers for SMEs: fragmented single market, regulatory complexity, limited access to capital, skills shortages, limited scale.
- Why it is our starting point: the handbook is a small, concrete contribution to exactly that diffusion problem.

Guided discussion

- Do you recognise the “Europe is behind” framing? Who makes the AI tools you named earlier?
- Draghi treats the productivity gap as a tech gap. What would closing it actually take for one local SME?
- If you were advising a regional SME tomorrow, what's the first thing they'd need to know about AI?

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Guided discussion, about 25 minutes. The slide carries three questions; there is a fourth to ask between the second and the third:

The report's lens is competitiveness. What does that lens underweight — trust, dependency on foreign vendors, jobs, data?

Take the first question back to the board from discussion 1: the tools they named, and who makes them.

The last question is the bridge into the assignments. End on it.

Teams of four

Four people, four AEL build roles, one team:

Product manager

Developer

Tester

Deployer

Everyone also researches in AIBS — same people build and research.

Sanity check: four people, four roles, at least one willing to drive the CLI — the command line used to direct the AI coding agent.

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Teams of four, formed now, together for the run. Each member takes one build role; everyone also researches.

Sanity check per team: four people, four roles, at least one willing to drive the command line, at least one who knows this region or can work in Dutch. If a team is missing one of the last two, say so now; it is far cheaper to fix in week 1 than in week 3.

Push for four people who are not already a group. A team that formed in the corridor tends to research the firm one of them already knew.

On roles: take the one you want to get better at. If you have never opened a terminal, developer is a legitimate choice.

AEL — Product Requirements Document

- **Problem and users.** What the platform is for, and who uses it — your research team, and, downstream, the SME owner.
- **What it must let a user do.** The path from raw source material to a published handbook page.
- **Output qualities.** What a good handbook page produced by the platform looks like.
- **Known constraints.** You direct an agentic CLI rather than hand-code; reuse existing infrastructure; a non-coder must be able to run it; every step needs a manual fallback.
- **Out of scope, for now.** What you are deliberately not building yet.
- **Open questions and assumptions.** What you're still unsure about.

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Hand out the PRD prompt. The platform helps the team research and produce the handbook, one page a week. Its user is the team itself; the SME owner is the downstream reader.

Walk the six sections. On constraints, say the four out loud: you direct an agentic CLI rather than hand-code; you reuse the wiki and the site; a non-coder must be able to run it; every automated step keeps a manual fallback.

Add the point the slide does not carry: much of the source material is Dutch — CBS, the Chamber of Commerce, sector bodies, the regional press, probably the interview too. Crossing that gap is a requirement of the platform, not a favour someone does by hand. Say so in the PRD.

The bar: a reader can tell what you are building, for whom, and how you will know it works.

AIBS — Research Proposal

- **Target audience.** Which SMEs is this for? Be specific — sector, size, region. A named partner company is ideal.
- **What they need.** What these SMEs need to know or decide about AI, and why — use the Draghi report and your own reasoning.
- **Research question.** One main question + at least 4 sub-questions — at least 2 on external analysis, at least 2 on internal analysis.
- **Quality criteria for the handbook.** What must it do to be genuinely useful to this audience?
- **Methodology.** Desk research + field research — one SME visit, weeks 2–4. Which questions does each method serve?

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Hand out the research proposal prompt. Walk the five sections.

On target audience, give the shape of "specific enough": a sector, a size, a region and a decision the owner is actually facing. The week 1 page has the worked example — metal and machine-building firms in the Achterhoek, 20 to 100 staff, spending two days a week turning drawings into quotations. It is there to show the shape, not to be copied.

Research question: one main question, at least four sub-questions, two external and two internal.

Quality criteria: this list becomes the checklist the handbook is later judged against. Methodology: desk research plus one SME visit in weeks 2 to 4, and which questions each serves.

1–3 pages each. This is a starting point.

- **PRD, the bar:** a reader can tell what you're building, for whom, and how you'll know it works.
- **Research proposal, the bar:** a reader can tell who the handbook serves, what it must answer, and how you'll judge it.

Push for a concrete target SME — sector, size, region — as early as possible.

You'll revise both as you learn more. Nothing here is final.

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The two documents constrain each other. The proposal's quality criteria and the PRD's output qualities describe the same thing from two sides. If they disagree, one of them is wrong; find out now, not in week 4.

Team work time, 2:25 to 3:40. Both lecturers circulate: AEL on the PRD, AIBS on the proposal. Push every team, early and repeatedly, to name a concrete target SME: sector, size, region.

Share-out and next steps

- Each team, one sentence: who is our handbook for?
- Each team, one sentence: what's an open question we're stuck on?

Next week is decided from what we see today.

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Share-out: each team, one sentence on who the handbook is for, and one open question they are stuck on.

Then the close, four points:

- Where the drafts go: post both in the team's Teams channel, due by the start of next session. The PRD is the first of six weekly AEL documents; the site lists them.
- The individual portfolio: one document per student, added to every week, 30 % of each mark; the interview carries the other 70 %. Say the four standard questions. Week 1's entry is shorter: what you contributed, what your role did first, and what you do not understand yet. Half a page.

- Two things the instrument lives or dies on: write it the same week, not at the end; and an honest small contribution scores better than an unsupportable large one, because the interview tests the claim.
- The two weekly gates are still not scored. Say it explicitly; students will assume otherwise now that something weekly carries a mark.

Next week is decided from what we saw today, partly from the "what I do not understand yet" answers.

After class: write the first decision-log entry, and read the portfolio entries.